Exercise series 2: ER to Relational Model

Introduction to databases

ADReM
Advanced Database Research & Modelling
University of Antwerp
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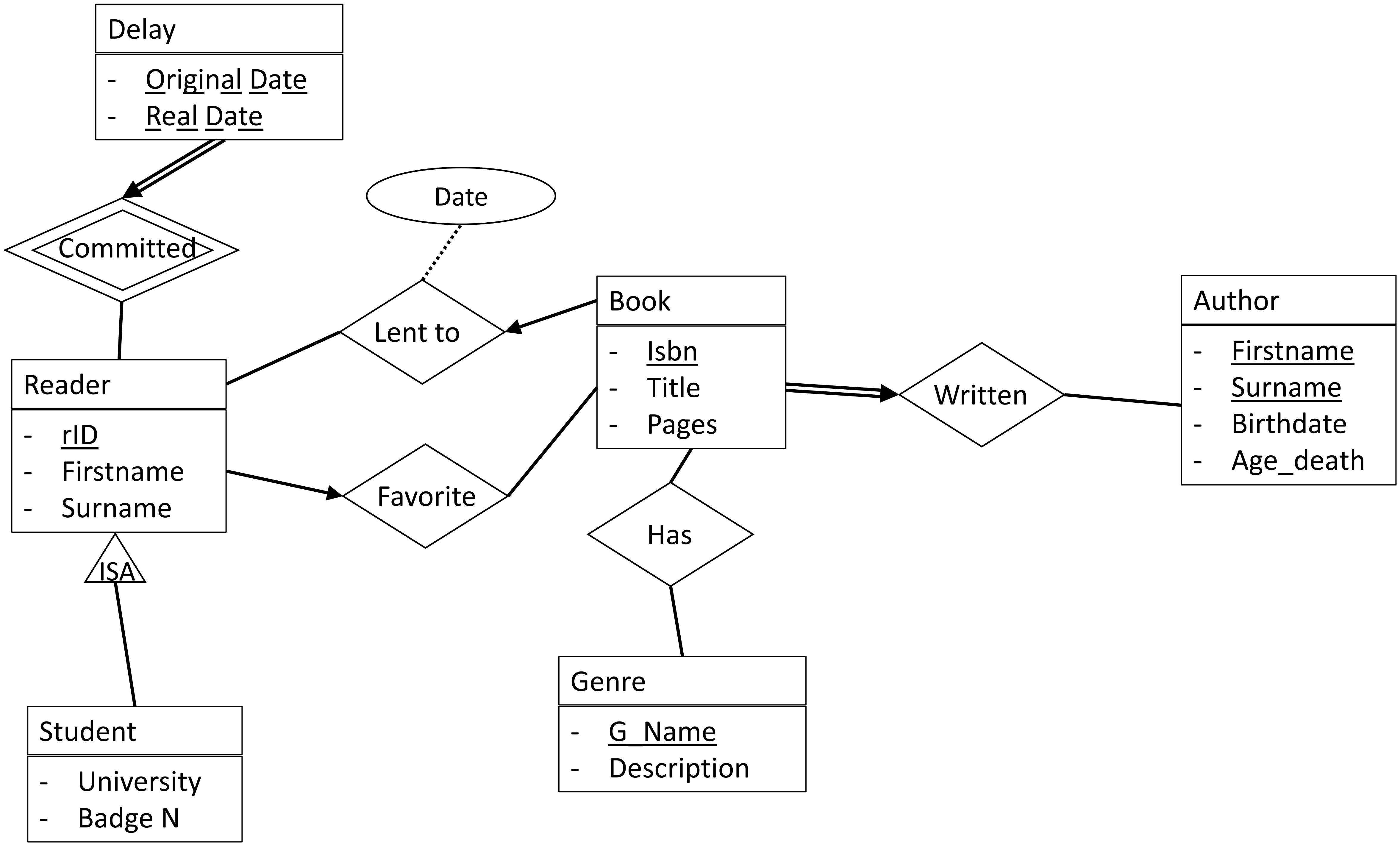
Starting ER Model



Tables for Entities

Try it on https://www.db-fiddle.com/

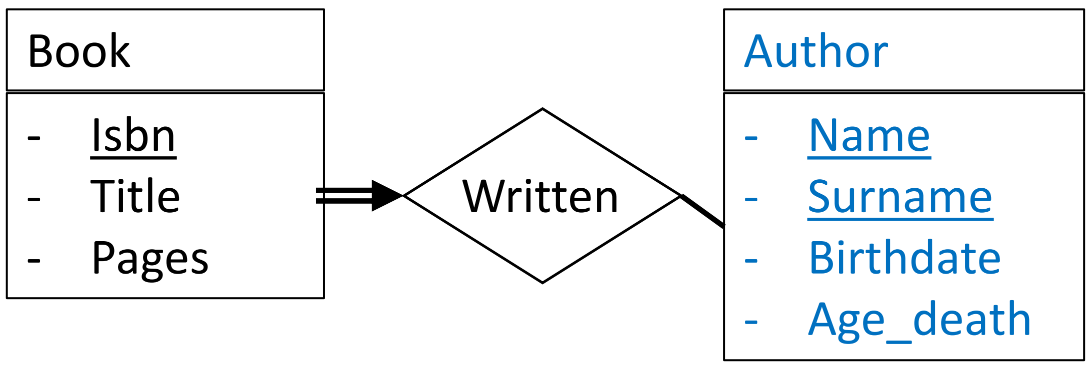
Author

- Firstname
- Surname
- Birthdate
- Age_death

```
CREATE TABLE Author (
   firstname VARCHAR(255) NOT NULL,
   surname VARCHAR(255) NOT NULL,
   birthdate VARCHAR(255) NOT NULL,
   age_death INT,
   PRIMARY KEY(name, surname)
);
```

- For each **entity** use `CREATE TABLE`
- Add one ore more **primitive attributes** and specify type: `INT, FLOAT, VARCHAR(n), DATE, TIME, TIMESTAMP`
- Add `NOT NULL` constraint to **mandatory** attributes
- Add `PRIMARY KEY`. The key is either a **single** attribute or **composed** of multiple attributes

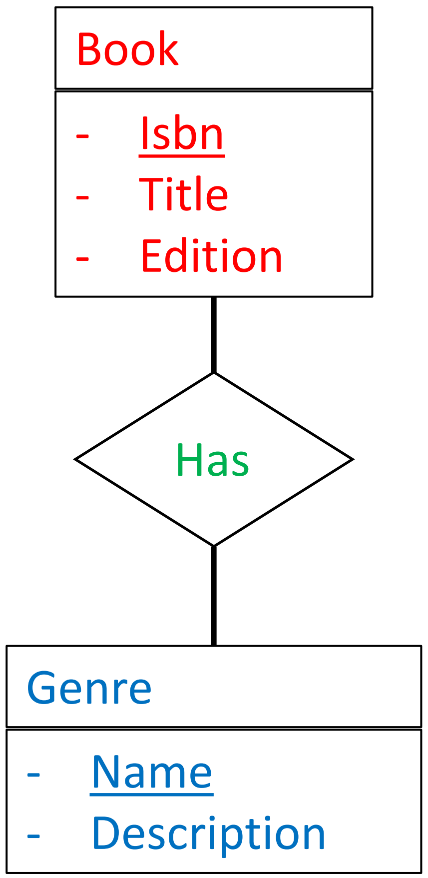
One to Many + Total Participation




```sql
CREATE TABLE Book(
    isbn VARCHAR(255) NOT NULL,
    title VARCHAR(255),
    pages INT,
    a_firstname VARCHAR(255) NOT NULL,
    a_surname VARCHAR(255) NOT NULL,
    FOREIGN KEY(a_firstname, a_surname)
    REFERENCES Author(firstname, surname),
    PRIMARY KEY(isbn)
    );
```

- Add a `FOREIGN KEY` reference in the new table
- Use key attribute(s) of the target table as `FOREIGN KEY` to have “one to many”
- Add `NOT NULL` constraint to have also “total participation”

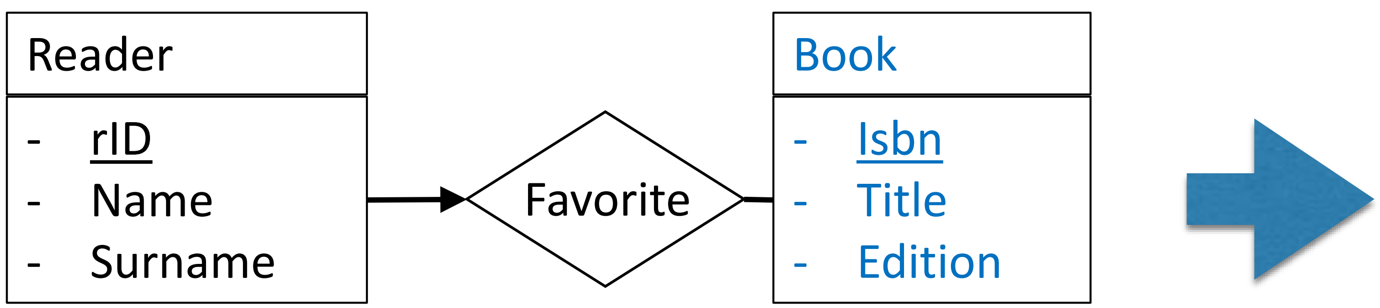
Many to Many




```
CREATE TABLE Has_genre (
    isbn VARCHAR(255) NOT NULL,
    FOREIGN KEY(isbn)
    REFERENCES Book(isbn),
    g_name VARCHAR(255) NOT NULL,
    FOREIGN KEY(g_name)
    REFERENCES Genre(g_name),
    PRIMARY KEY(b_isbn , g_name)
    );
```

- For each **many-to-many relation** use `CREATE TABLE`
- Add key attribute(s) of first table and add `FOREIGN KEY`
- Add key attribute(s) of second table and add `FOREIGN KEY`
- `PRIMARY KEY` is composed of key attributes of both sides

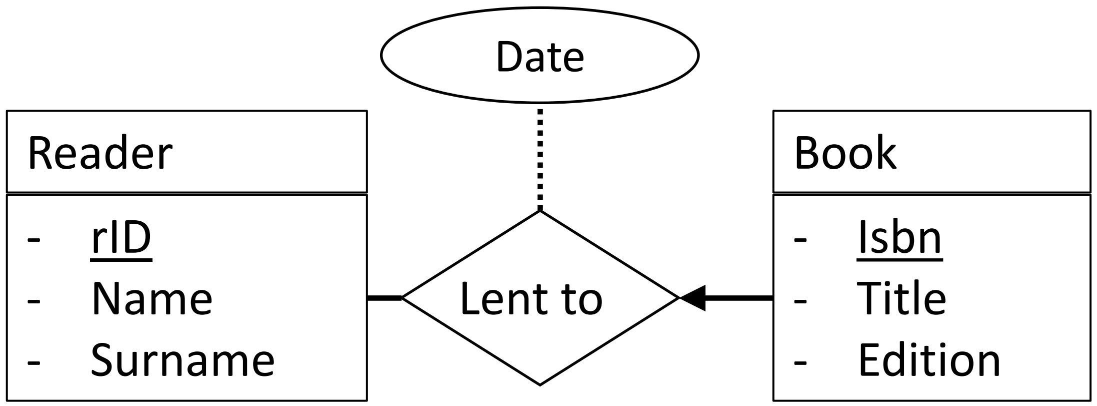
One to Many (no extra table)




```
CREATE TABLE Reader(
    rid INT NOT NULL,
    firstname VARCHAR(255) NOT NULL,
    surname VARCHAR(255) NOT NULL,
    favorite_b VARCHAR(255),
    FOREIGN KEY(favorite_b)
    REFERENCES Book(isbn),
    PRIMARY KEY(rid)
    );
```

- Add a `FOREIGN KEY` reference in the new table
- Use key attribute(s) of the target table as `FOREIGN KEY` to have "one to many"
- **DO NOT** add `NOT NULL` constraint to foreign key attributes!

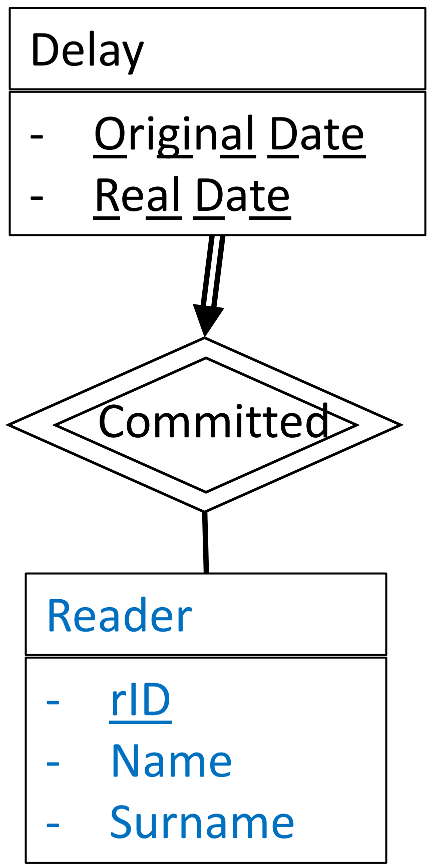
One to Many (with extra table)




```
CREATE TABLE Lent_to(
    rid INT NOT NULL,
    isbn VARCHAR(255) NOT NULL,
    borrowing_date DATE,
    FOREIGN KEY(rid)
    REFERENCES Reader(rid),
    FOREIGN KEY(isbn)
    REFERENCES Book(isbn),
    PRIMARY KEY(isbn)
    );
```

- Add key attribute(s) of first table and add `FOREIGN KEY`
- Add key attribute(s) of second table and add `FOREIGN KEY`
- `PRIMARY KEY` is composed of key attributes of JUST ONE SIDE

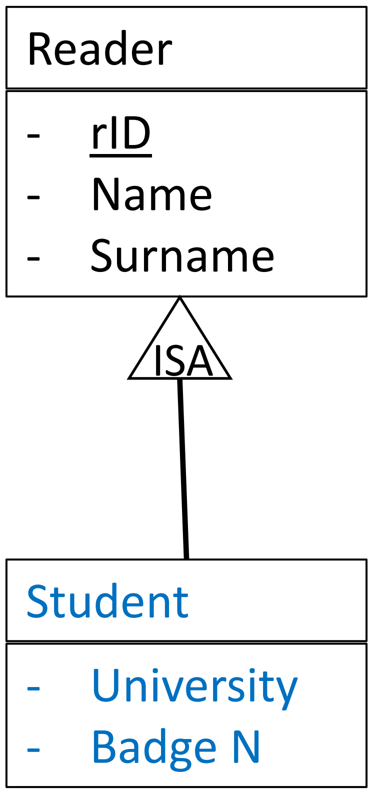
Weak Entity Set




```
CREATE TABLE Delay(
    rid INT NOT NULL,
    original_date VARCHAR(255) NOT NULL,
    real_date VARCHAR(255) NOT NULL,
    FOREIGN KEY(rid )
    REFERENCES Reader(rid),
    PRIMARY KEY(rid, original_date, real_date)
    );
```

- Add a `FOREIGN KEY` reference in the new table
- Use key attribute(s) of the target table as `FOREIGN KEY` to have “one to many”
- Add `NOT NULL` constraint to have also “total participation”
- Use `FOREIGN KEY` attribute(s) as part of table primary key

ISA Relationship




```sql
CREATE TABLE Student(
    rid INT NOT NULL,
    university VARCHAR(255) NOT NULL,
    badge_n INT NOT NULL,
    FOREIGN KEY(rid)
    REFERENCES Reader(rid),
    PRIMARY KEY(rid)
    );
```

- Add a `FOREIGN KEY` reference in the new table
- Use key attribute(s) of the target table as `FOREIGN KEY` to have “one to many”
- Add `NOT NULL` constraint to have also “total participation”
- Use `FOREIGN KEY` attribute(s) as primary key

Extra SQL syntax

- **INSERT INTO** tablename(cols) **VALUES**(val_1,…, val_k);
- **UPDATE** tablename **SET** col1=val1 **WHERE** col2=val2;
- **DELETE FROM** tablename **WHERE** col2=val2;

Examples:

```
INSERT INTO Author(name, surname) VALUES ('jk','rowling');

UPDATE Reader SET favorite_b='8601402233168' WHERE rid=2;

DELETE FROM Student
WHERE university = 'uantwerpen' AND badge_n = 123456;
```